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Indian Ocean SST and Dipole Outlook

Forecast issued 31 August 2026 · covering 1–30 September 2026

ACCORD / CHC — draft for review

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Abstract

This is a forecast of sea surface temperature in the two Indian Ocean Dipole boxes, and of the dipole itself, for the next 30 days and for each of the next four weeks. It uses the ECMWF sub-seasonal (S2S) forecast, corrected against NOAA OI SST by linear regression.

The forecast is for a positive Indian Ocean Dipole at every horizon, of weak-to-moderate amplitude: $+0.43$ °C over the first week, about $+0.82$ to $+0.86$ °C in weeks 2 and 3, and $+0.72$ °C averaged over the next 30 days. This is driven almost entirely by a warm western box, forecast at 28.4 – 28.8 °C, 0.7 – 1.0 °C above its 2006–2025 average for these dates. The eastern box is close to normal, 0.1 – 0.3 °C above.

A positive IOD developing through spring is the broad consensus — the Bureau of Meteorology, JAMSTEC, IRI and the WMO Global Seasonal Climate Update all forecast one. But **our near-term values sit above the independent observed estimates**, and this should be weighed before acting on the first week. The Bureau's index for the same week is $+0.25$ and it calls the IOD *neutral*, having been below the $+0.4$ threshold for five consecutive weeks with forecast magnitudes easing. Computed on our own data and baseline, the observed dipole for that week is $+0.01$ against our forecast $+0.43$ — inside the stated interval, but at its edge (see *Early verification*).

Only at week 2 does the forecast measurably beat persistence — and there it does so in both boxes. Against a 30-day persistence baseline the week-2 gain in correlation is $+0.16$ (west) and $+0.15$ (east), and in both cases an 80% bootstrap interval on the gain excludes zero. At every other horizon the gain is **not separable from sampling noise on twenty years**: the eastern box gains $+0.21$ at week 3 and $+0.18$ at week 4, but those intervals straddle zero, as does the $+0.11$ on the 30-day window. This applies symmetrically — the small negative gains at week 1 and at week 4 in the west are equally inconclusive. Twenty years is simply too short to resolve differences of this size.

Method

Forecast model. ECMWF S2S extended-range ensemble, issued 31 August 2026: 100 members, 46 daily steps, 1.5° grid. From the ECMWF Data Store (ecds.ecmwf.int), dataset `s2s-forecasts`, variable `sea_surface_temperature`.

Training data. The reforecast set for the same calendar date (`s2s-reforecasts`) — the same run for 31 August in each of the twenty years 2006–2025, 10 members each. The correction is fitted on these twenty past forecasts.

Observations. NOAA OI SST v2.1, 0.25° daily, from NOAA PSL — the record CHC uses.

Boxes. Saji et al. (1999) dipole boxes. West 50 – 70 °E, 10 °S– 10 °N; east 90 – 110 °E, 10 °S– 0 . Averages are area-weighted by cosine of latitude on each dataset's own grid.

Lead windows. Week 1 is forecast days 1–7, then 8–14, 15–21, 22–28, and 1–30. S2S SST is filed as a 24-hour mean, so day 1 is the **day of the 00Z issuance**, not the day after — week 1 therefore verifies 31 August to 6 September. The 30-day window includes the issuance day, whose ocean state is already partly known.

Correction. For each box and window separately, a straight line fitted by ordinary least squares — observed temperature against forecast temperature across the twenty reforecast years — then applied to this year's forecast. Regression rather than quantile matching, because a straight line continues a warming trend outside the historical range while quantile matching pulls values back inside it.

Uncertainty. The 80% figure is a prediction interval on Student's t with 18 degrees of freedom, carrying both the uncertainty in the fitted line and the residual scatter. It is deliberately not the naive $1.28 \times \text{residual sd}$: **this forecast sits 2.3–3.0 standard deviations above the training mean in the western box**, so the extrapolation term is substantial, and ignoring it understated the western interval by 20–28%. The interval reflects how wrong this method has been on past years — not how much this week's 100 members disagree with each other.

Testing. All accuracy figures are leave-one-year-out: the line is refitted without each year, then used to predict it, so nothing quoted is measured on data used to fit it.

Reproducing. `iod_pipeline.run("2026-08-31")` in `experiments/IOD-forecasting/`; the date is the only input. Reforecasts exist only for Monday and Thursday issuances, so those are the valid init dates. Data via `acmadDL`, analysis via `africas2s`.

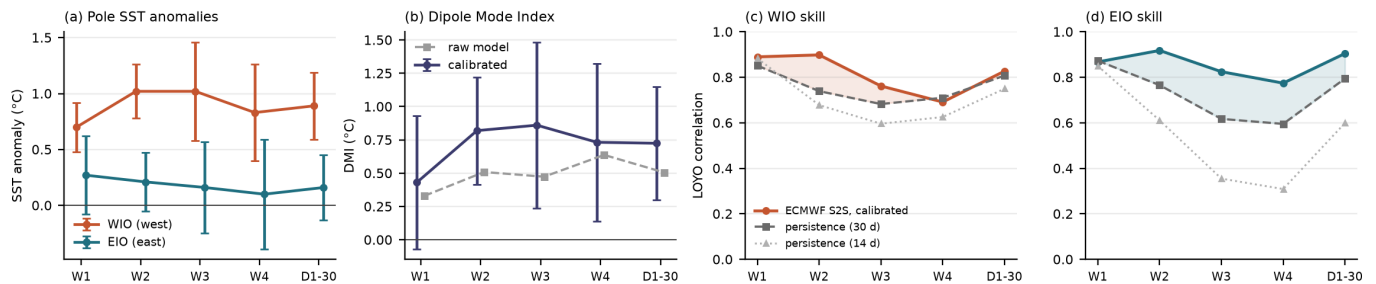


Figure 1: **(a)** Forecast temperature anomaly in each box, with 80% prediction intervals. **(b)** The dipole index, calibrated and raw — the gap between them is the regression's amplitude correction, not bias removal. **(c,d)** Leave-one-year-out correlation over twenty past years against two persistence baselines; the shaded gap is what the model adds over the harder 30-day one.

Results

Horizon	Valid dates	West box	East box	Dipole
Week 1	31 Aug – 6 Sep	28.36 °C (+0.70)	28.36 °C (+0.27)	+0.43 ± 0.50
Week 2	7–13 Sep	28.73 °C (+1.03)	28.31 °C (+0.21)	+0.82 ± 0.40
Week 3	14–20 Sep	28.81 °C (+1.02)	28.26 °C (+0.16)	+0.86 ± 0.62
Week 4	21–27 Sep	28.78 °C (+0.83)	28.28 °C (+0.10)	+0.73 ± 0.59
Days 1–30	31 Aug – 29 Sep	28.68 °C (+0.89)	28.29 °C (+0.17)	+0.72 ± 0.42

Temperatures are absolute; brackets are the departure from the **2006–2025** average for the same dates, dipole in °C with its 80% interval. That baseline is the reforecast period, not the 1991–2020 climatology BoM, CPC and NOAA use — so **these anomalies are not directly comparable with the dipole values those centres publish**.

The raw model dipole is smaller (+0.33 to +0.64). The difference between the raw and calibrated columns is **the regression's amplitude correction, not bias removal**: least squares with an intercept preserves the mean, so any constant bias cancels on both sides. What the calibration changes is amplitude — it stretches the western box (slope 1.04–1.42, i.e. the ensemble mean under-disperses it) and shrinks the eastern one (slope 0.62–0.80, where the model over-amplifies variability). The calibrated dipole is the product; the raw column is shown only so the size of that correction is visible.

Accuracy

Leave-one-year-out correlation over the twenty reforecast years, against persistence — “the ocean stays as it is at issue”. Persistence is a hard baseline for SST, and how hard depends on how long a window it averages, so both are shown.

	Week 1	Week 2	Week 3	Week 4	Days 1–30
West — model	0.89	0.90	0.76	0.69	0.83

	Week 1	Week 2	Week 3	Week 4	Days 1–30
West — persistence, 30 d / 14 d	0.85 / 0.88	0.74 / 0.68	0.68 / 0.60	0.71 / 0.63	0.81 / 0.75
West — gain (80% interval)	+0.04 (–0.04, +0.12)	+0.16 (+0.02, +0.28)	+0.08 (–0.07, +0.22)	–0.02 (–0.11, +0.14)	+0.02 (–0.07, +0.14)
East — model	0.87	0.92	0.83	0.77	0.90
East — persistence, 30 d / 14 d	0.87 / 0.85	0.77 / 0.61	0.62 / 0.36	0.60 / 0.31	0.79 / 0.60
East — gain (80% interval)	–0.01 (–0.06, +0.04)	+0.15 (+0.03, +0.20)	+0.21 (–0.02, +0.32)	+0.18 (–0.05, +0.30)	+0.11 (–0.02, +0.19)

Two things govern how this table should be read.

The baseline length changes the answer. An earlier draft of this report used only the 14-day baseline and claimed the forecast beat persistence at every horizon with a margin that grew with lead. That is true only for that one window — the weakest of the three tested. It does not survive the 30-day comparison, which is the more natural benchmark for a product whose flagship window is a 30-day mean, and it has been withdrawn.

Twenty years cannot resolve most of these differences. The gain rows above carry an 80% bootstrap interval, resampling the (model, persistence, observed) triples so the dependence between the two predictors is preserved.

Week 2 is the only horizon where the gain excludes zero, and it does so in both boxes. Weeks 3 and 4 in the east have the largest point gains in the table and still cannot be separated from noise; so can the 30-day window, which is the horizon most likely to be quoted. The same standard applies to the negative numbers: week 1, and week 4 in the west, are inconclusive rather than demonstrated failures.

Typical error of the corrected forecast is 0.15–0.29 °C (west) and 0.20–0.37 °C (east).

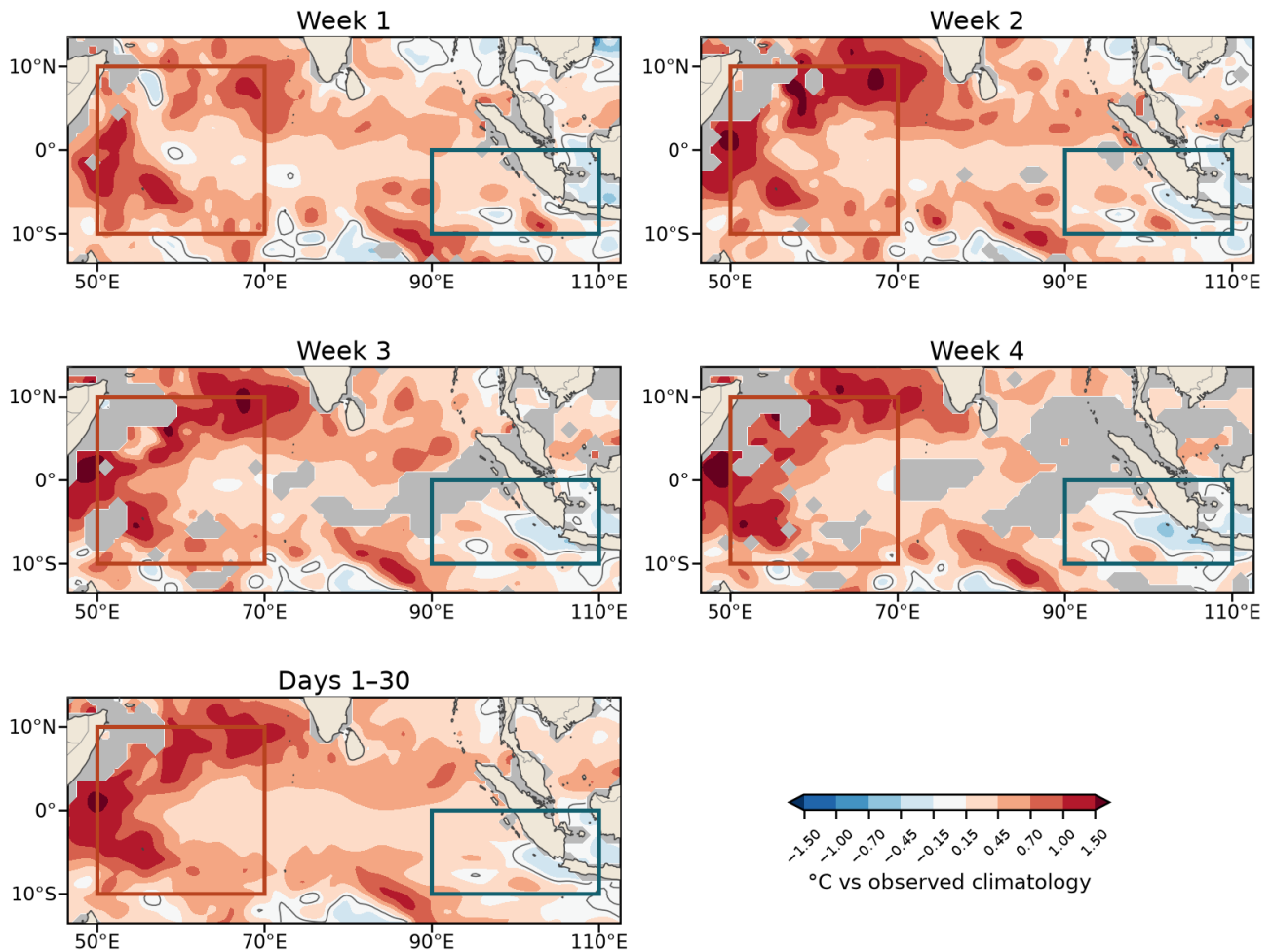


Figure 2: Corrected SST anomaly at every horizon (°C vs the 2006–2025 observed average). Boxes: western (orange) and eastern (teal) poles; land in tan. Grey ocean is where leave-one-year-out correlation falls below 0.4 and the forecast should not be relied on. That area expands with lead across the basin, and the eastern box is consistently the more affected of the two. Each cell is corrected by its own regression; observations are averaged from 0.25° to the model's 1.5° grid. Fields are cubic-refined for contouring — presentational only, and the zero contour in particular is placed by the interpolation rather than measured; every number quoted above comes from the native 1.5° field.

Early verification

OISST is already available for the whole of the week-1 window, so that forecast can be checked now.

	forecast (calibrated)	forecast (raw)	observed	error
West	28.36	28.47	28.23	+0.13 (interval ± 0.22)
East	28.36	28.65	28.65	-0.29 (interval ± 0.36)
Dipole	+0.43	+0.33	+0.01	+0.42 (interval ± 0.50)

Every error is inside its 80% interval, but two things deserve saying plainly. The dipole error is close to the edge of the interval, and it is in the direction of the forecast being **too positive** — consistent with BoM calling the IOD neutral for the same week. And in the eastern box the **raw model was essentially exact (28.65 against 28.65 observed) while the calibration moved it 0.29 °C the wrong way**. That is the shrinkage risk of a twenty-point fit made concrete: the slope of 0.80 is estimated from twenty years and cannot know when a given year does not need shrinking. One week is not a verification, but it is a reason to treat the eastern calibration with some caution and to keep verifying.

Limitations

- **Only week 2 shows a gain over persistence that twenty years can resolve.** Weeks 3–4 and the 30-day window have larger point gains in the east but intervals straddling zero. Week 1 is indistinguishable from persistence rather than worse than it.
- **Near-term values run above independent estimates.** BoM has the IOD neutral (+0.25) for the week we forecast at +0.43; our own observations give +0.01 for that week. The disagreement on *direction over the season* is nil — everyone forecasts a positive IOD — but the near-term level should be treated as an upper estimate.
- **The baseline is 2006–2025, not 1991–2020.** Anomalies and the dipole here are not directly comparable with BoM, CPC or NOAA published values, which also use different SST analyses.
- **Twenty years is a short training record,** and this forecast extrapolates 2.3–3.0 standard deviations beyond the training mean in the west. The interval accounts for that; the slope estimate itself still rests on twenty points.
- **The interval is historical,** not ensemble-based. A version using the 100 members' spread is the obvious next step.
- **Map and box numbers differ by 0.15–0.35 °C** — correcting each cell then averaging is not the same calculation as correcting the box average; quote the table for a box.
- **Box footprints differ slightly between model and observations** (cell-centre masking on a 1.5° grid versus 0.25°), so part of the model–observation offset is footprint rather than model error. The intercept absorbs the constant part.
- **The dipole is built from separately calibrated poles, and that choice is worth ~0.4 °C.** Calibrating each pole and differencing multiplies the raw dipole by 1.15–1.82×. Regressing the observed dipole directly on the model dipole instead gives shrinkage, and week-2 forecasts of +0.41 against our +0.82. Our construction verifies better out-of-sample and is what Funk asked for (poles separately, then combined), but a choice that moves the headline by more than the +0.4 °C threshold itself deserves stating.
- **Persistence is given a one-day advantage.** Its averaging window ends on the init day, which is also day 1 of the verification window, and assumes same-day observations that would not be available at issue. Both make persistence look better, so the comparisons above are conservative.
- **One model only,** ECMWF S2S. GEFSv12 was considered and set aside.